Benjamin Seguin

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Education

Université PSL — ENS, Dauphine, Mines Paris, France
M.Sc. in Computer Science (IASD track) *Sep. 2025 – Sep. 2026*

- Relevant coursework: Mathematics of Deep Learning (B. Loureiro), Dimension Reduction and Manifold Learning (E. Aamari), Non-convex Inverse Problems (I. Waldspurger), Foundations of Machine Learning (F. Bach), Optimization for Machine Learning (C. Royer).

HEC Montréal Montréal, Canada
M.Sc. in Financial Engineering, Thesis track *Sep. 2023 – Dec. 2024*

- GPA 4.0/4.3; *mention d'excellence*; Excellence Scholarship.
- Quantitative coursework: Stochastic Calculus (A), Machine Learning for Financial Data (A+), Time Series Econometrics (A+), Statistical Methods for Financial Data (A).
- Master's thesis (4.3/4.3, *mention d'excellence*): *Strategies to Monetize Sentiment Extracted from Earnings Call Transcripts Using NLP and LLMs.*
- Teaching Assistant in Time Series Econometrics.

HEC Montréal Montréal, Canada
B.B.A. in Finance, Bilingual track *Sep. 2018 – Jun. 2022*

Research Experience

LAMSADE, MILES team — Université Paris Dauphine-PSL Paris, France
Research Intern, supervised by Prof. Clément Royer *Apr. 2026 – Sep. 2026*

- Complexity analysis of a hybrid algorithm combining Nonlinear Conjugate Gradient with Negative Curvature Descent for non-convex optimization. Derived per-regime iteration complexity bounds on strict saddle functions, including novel rates in the local strong convexity regime.

HEC Montréal — Computational Finance Montréal, Canada
Research Assistant, supervised by Prof. David Ardia *Jan. 2025 – Sep. 2025*

- Developed three open-source R packages for econometric inference (regime-switching correlations, mean-variance spanning tests, luck-corrected peer performance). Three co-authored preprints on SSRN (Ardia & Seguin, 2025).

Professional Experience

Caisse de Dépôt et Placement du Québec (CDPQ) Montréal, Canada
Quantitative Research Intern *Jun. 2024 – Dec. 2024*

- Built ML pipelines for LLM-based sentiment extraction (reduced inference time 25min → 2min); developed OOP architecture for low-latency quantitative signal computation.

Technical Skills

- **Programming:** Python (PyTorch, JAX, NumPy, Pandas, scikit-learn), R, LaTeX, Git.
- **Mathematics:** Linear algebra, optimization, probability, stochastic calculus, time series.
- **Languages:** French (native), English (fluent).